

TITLE PAGE

- Problem Statement ID - **SIH25072**
- Problem Statement Title - **Design and Implementation of Solar-Powered Dewatering in Mining Operations.**
- Theme - **Renewable / Sustainable Energy**
- PS Category- **Hardware**
- Team ID -
- Team Name - **RESISTOR**

PUMPED HYDRO AND TRAIN BASED ENERGY STORAGE SYSTEM TO INCREASE THE EFFICIENCY OF COPPER MINE DEWATERING PUMP SYSTEM

DETAILED EXPLANATION OF THE PROPOSED SOLUTION

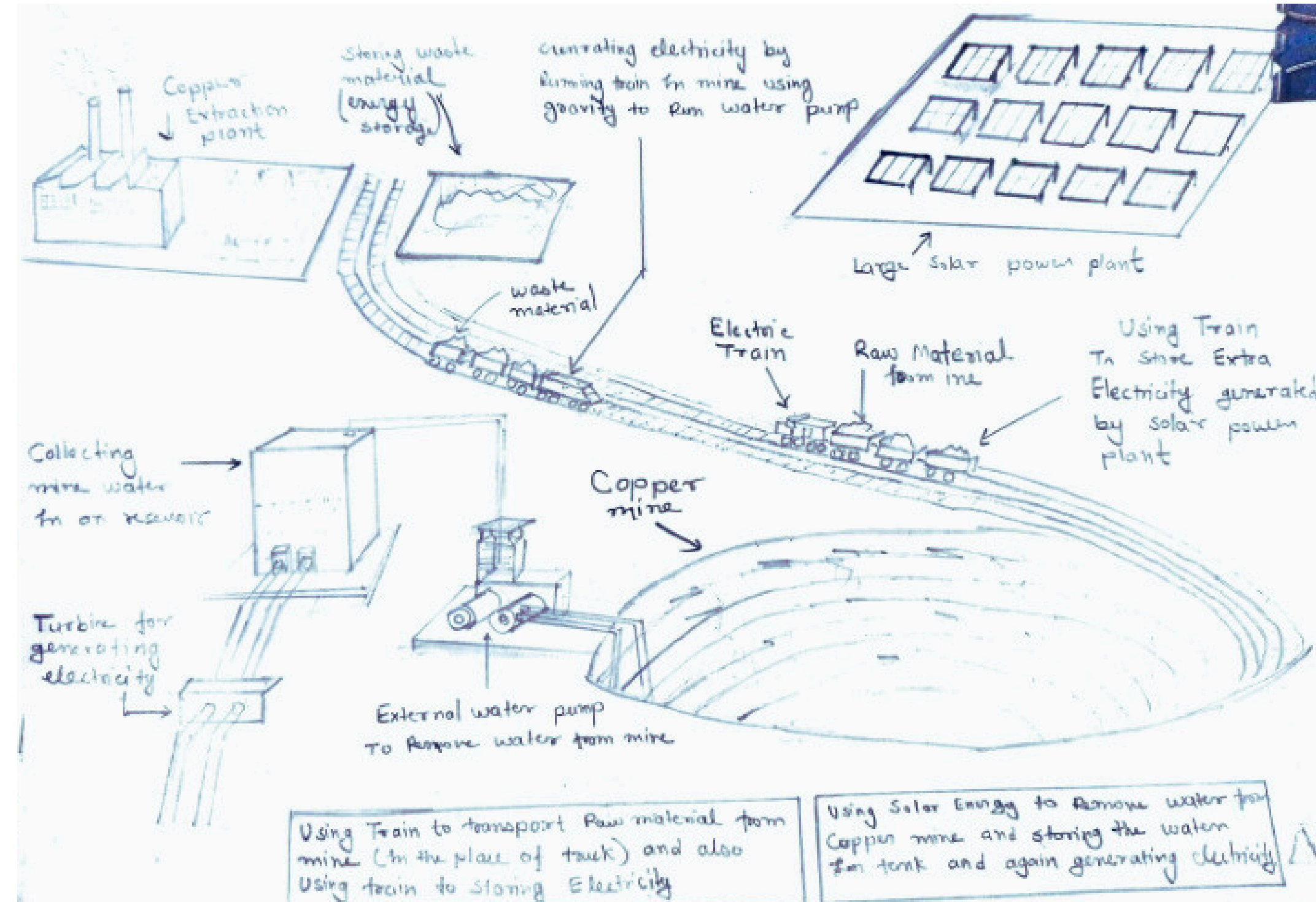
Our pumped hydro and train-based energy storage systems store the excess electricity produced by the solar power plant during the daytime. This stored energy can be used to operate the dewatering system of the copper mine at night or whenever needed. By doing so, we can reduce carbon emissions caused by diesel-operated pumps and make the process more cost-effective

HOW IT ADDRESS THE PROBLEM

Storing the excess electricity produced by the solar plant during the daytime and using it at night to run the water pump helps increase efficiency

INNOVATION AND UNIQUENESS OF THE SOLUTION

The innovation of this idea lies in using stored solar energy to power the dewatering system in copper mines, reducing reliance on diesel pumps. Its uniqueness comes from integrating renewable energy storage with mining operations, making the process more efficient, cost-effective, and environmentally friendly.

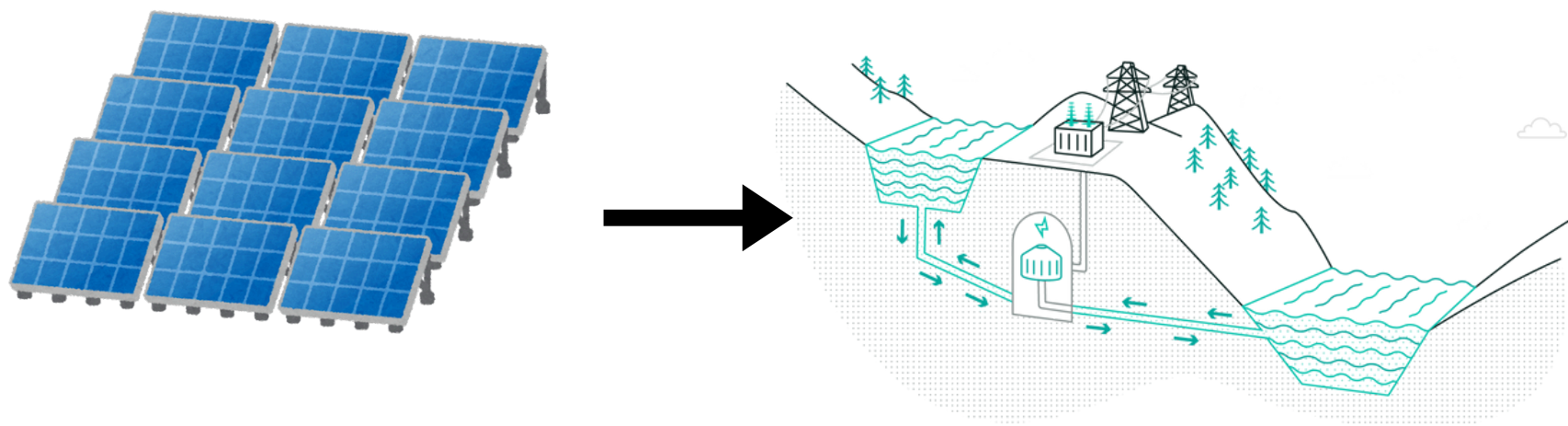


TECHNOLOGIES TO BE USED (E.G. PROGRAMMING LANGUAGES, FRAMEWORKS, HARDWARE)

We are using simple and readily available hardware to increase the efficiency of the dewatering pumps in copper mines. Specifically, we use a pumped hydro system that stores energy by utilizing copper mine water, and a traditional train-based system that stores electricity during the daytime. At night, this stored energy is supplied to the dewatering pumps, while the same train system is also used for transporting materials from the mine

METHODOLOGY AND PROCESS FOR IMPLEMENTATION (FLOW CHARTS/IMAGES/WORKING PROTOTYPE)

The process involves storing excess electricity generated during the day and using it at night to power the dewatering pumps in copper mines. Water from the mine is pumped to a higher elevation using surplus energy and later released to generate electricity when needed. Similarly, trains are moved uphill during the day to store energy, and their downhill movement at night is used to recover electricity. This method ensures efficient energy use, supports continuous pump operation, and reduces reliance on diesel fuel



DAY TIME (CHARGING)



NIGHT TIME (DISCHARGING)

ANALYSIS OF THE FEASIBILITY OF THE IDEA

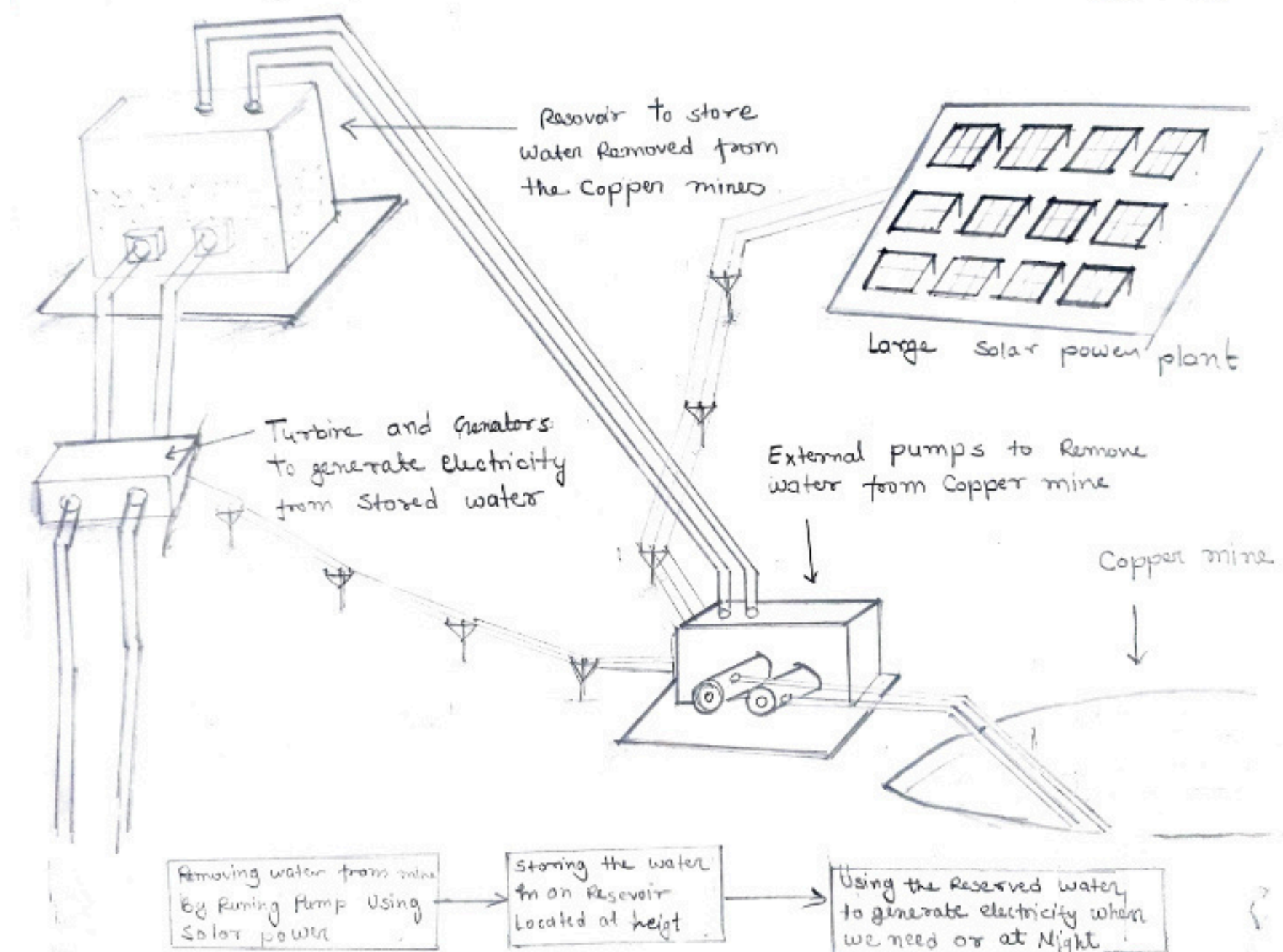
This method is feasible as it utilizes existing copper mine processes, like water management and material transport, to store and use energy efficiently. By using surplus solar electricity during the day, it reduces infrastructure needs and lowers operational costs. The system integrates easily with current mining operations, reduces fuel consumption and carbon emissions, and is both environmentally and economically sustainable. The availability of solar energy and the scale of mining operations further support its practicality.

POTENTIAL CHALLENGES AND RISKS

The main challenges are variability in solar energy, energy losses during storage, coordination of operations, and ensuring safety and system reliability. Proper planning and maintenance are required to address these risks.

STRATEGIES FOR OVERCOMING THESE CHALLENGES

To address these challenges, energy management systems can be used to optimize storage and usage. Backup power sources can support operations during low solar availability. Regular maintenance and monitoring ensure safety and reliability, while efficient system design helps reduce energy losses.



POTENTIAL IMPACT ON THE TARGET AUDIENCE

This method benefits copper mine operators by reducing fuel costs and improving energy efficiency. It also helps workers by providing a more reliable and cleaner power source, minimizing disruptions. Additionally, it supports environmental goals by lowering carbon emissions, contributing to a healthier workplace and surrounding community.

BENEFITS OF THE SOLUTION (SOCIAL, ECONOMIC, ENVIRONMENTAL, ETC.)

- Economic: Reduces operational costs by lowering fuel consumption and using surplus solar energy efficiently.
- Environmental: Decreases carbon emissions and reliance on diesel, supporting cleaner and sustainable mining operations.
- Social: Provides a safer and more reliable power supply, improving working conditions and reducing health risks associated with pollution.
- Operational: Enhances energy efficiency, ensures continuous dewatering pump operation, and supports better resource management.

RESEARCH AND REFERENCES
